Bachelor of Science (Honours) in Data Science and Artificial Intelligence

DA 102 – Excel Overview

Learning Objectives

01 What are cells and their capabilities?

02 What is a formula?

03 What are functions? How to use them

04 Understand logical functions

05 Understand references

06 Understand errors

Cells in Excel

Cells

- A fundamental unit in Excel.
- It has location referred by a pair (Column name, row number). Example: A1, B7, D12
- It hold various data types such as numbers, real valued numbers, text, dates, etc.
- Cells are used to perform desired computations.
- Cells can be formatted to change their appearance. Example, increase font size, use of font type, color, and alignment.

Data in a Cell

Data in a Cell - 01

- A cell can hold variety of data or objects
- Numbers, and real numbers
- Text
- Date & time
- Formulas
- Functions

Data in a Cell - 02

- Hyperlinks
- Images and objects
- Notes and/or comments
- Validation rules
- Symbols and special characters
- Rich text formatting

Hyperlinks

- A website address
- An email address
- Files: a hyperlink to a PDF document
- A cell within a workbook
- A place in this document
- External workbooks that reside on the very computer you are working
- Online documents such as a document in google drive
- Online maps and resources

Referring To Group Of Contiguous Cells

Referring Contiguous Cells

- Row selection notation: C1:C7 refers to Cells C1, C2, C3, C4, C5, C6 and C7
- Column selection notation: A1:F1 refers to Cells A1, B1, C1, D1, E1 and F1
- Rows and columns selection notation: A1:C3 refers to group of cells A1, A2, A3, B1, B2, B3, C1, C2 and C3.
- The COLON notation is used in formulas and functions extensively.
- Group of cells can also be accessed through Graphical User Interface methods using mouse, keyboard or a combination.
- Mouse: Left click on cell C1. Hold the left button of the mouse and drag the mouse pointer till cell C7.

Referring Contiguous Cells

- Keyboard:
- Click on cell C1.
- Hold down the "Shift" key on your keyboard.
- Use the "Down arrow" key to extend the selection till cell C7 is reached.
- All cells between C1 and cell C7 will be selected

Referring Contiguous Cells

- **Keyboard & Mouse:**
- **Click on cell C1.**
- **Hold down the "Shift" key on your keyboard.**
- **Click on cell C7.**
- **All cells between cell C1 and cell C7 will be selected.**

Five Important Elements

Five important Elements

- Formulas
- Functions
- Logical functions
- References
- Errors

Formulas

Overview of Formulas

- Formula specify the computations to be performed
- An expression involving constants or cells or their combination
- Output: Result of the computation stored in a separate cell

Example formula involving constant expressions

Example formula involving two cells

Example formula involving more than two cells

Example formulas and non-contiguous cells

Example formulas horizontal cells

Displaying formulas

Copying formula to same row but varying column

Copying Formulas – Same Row, Varying Column

- Formula in Cell D2 = B2 + C2.
- We copy formula in cell D2 to cell G2.
- There is no change in the target row where the formula is being copied (G2)
- D2 contains two operands B2, C2
- G2 also contains two operands ?2, ?2
- ? Will be dynamically computed while performing copying

Formulas – Observations – Computation of First Term in G2

- First term of the formula in D2 is B2. Column B is two columns to the left of D
- In the target cell G2, compute two columns to the left of G.
- Column E is two columns to the left of G.
- Therefore, first term in G2 is E2.

Formulas – Observations – Computation of Second Term in G2

- Second term of the formula in D2 is C2. Column C is one column to the left of D
- In the target cell G2, compute one column to the left of G.
- Column F is one column to the left of G.
- Therefore, second term in G2 is F2.
- Copy of result in G2 is: E2 + F2

Copying formula to same column but varying row

Copying Formulas – Same Column, Varying Row

- Formula in Cell B4 = B2 + B3
- We copy formula in cell B4 to cell B7.
- There is no change in the target column where the formula is being copied (B7)
- B4 contains two operands B2, B3
- B7 also contains two operands B?, B?
- ? Will be dynamically computed while performing copying

Formulas – Observations – Computation of First Term in B7

- First term of the formula in B4 is B2. Row 2 is two rows above the source formula row 4.

- In the target cell B7 compute two rows above the target row 7.

- Row 5 is two row above target formula row (7).

- Therefore, first term in B7 is B5.

Formulas – Observations – Computation of Second Term in B7.

- Second term of the formula in B4 is B3. Row 3 is one row above the source formula row 4.
- In the target cell B7 compute one row above the target row 7.
- Row 6 is one row above target formula row (7).
- Therefore, second term in B7 is B6.
- The result of copying formula in cell B4 to cell B7: B5 + B6

Moving formulas

Formulas – Matrix Operations

Matrix Operations

- Given two matrices M1 and M2 of size 5 x 5 each
- Write formulas to perform matrix addition
- Write formulas to perform matrix subtraction
- Write formulas to perform element-wise matrix multiplication
- Write formulas to perform standard matrix multiplication – with present understanding we must hand compute the result by writing 25 formulas.

Functions

Functions

- A function is a pre-built formula which performs an already identified computation.
- Function must be written as a formula within a cell.
- Function takes one or more inputs. Also known as arguments.
- Returns an output in the cell in which it got invoked.
- The number of arguments a function takes depends on which function is being invoked.

Categories of functions

- Mathematical Functions
- Text Functions
- Logical Functions
- Date and Time Functions
- Financial Functions
- Statistical Functions
- Lookup and Reference Functions
- Array Functions

Mathematical Function Examples

SUM Function

- SUM function accepts one or more numbers
- SUM function accepts one or more cells
- SUM function accepts one or more group of cells
- SUM function accepts any combination of the above

AVERAGE Function

- AVERAGE function accepts one or more numbers
- AVERAGE function accepts one or more cells
- AVERAGE function accepts one or more group of cells
- AVERAGE function accepts any combination of the above

SUMPRODUCT Function

Item	Quantity	Price	SUMPRODUCT
Product A	10	5	=10 * 5 + 15 * 7.5 + 20 * 10 =362.5
Product B	15	7.5	
Product C	20	10	

Logical Functions

IF Function

- IF allows to make logical comparison between two values or two cells
- IF function takes three arguments
- First argument is the comparison involving two values or two cells.
- Second argument is the result of IF function when the comparison is TRUE
- Third argument is the result of IF function when the comparison is FALSE.
- First argument is also known as testing condition

Examples

AND Function

AND Function

- **AND function allows to check whether ALL conditions specified are TRUE or NOT.**
- **AND function takes one or more conditions as arguments.**
- **AND returns TRUE when every condition results in TRUE.**
- **Among the arguments, even if one argument result in FALSE, the AND will return FALSE.**

OR Function

OR Function

- OR function allows to check whether ANY one of the specified conditions is TRUE or NOT.
- OR function takes one or more conditions as arguments.
- OR returns TRUE when one among the given conditions results in TRUE.
- When every argument is evaluated to FALSE, OR function result in FALSE.

Reference Functions

Reference Functions

- These are category of functions that allows us to work with
  - Cell references
  - Cell ranges and
  - Addresses
- These class of functions are used to extract information about
  - Cells and ranges
  - Manipulate cell references
  - Perform computations based on cell location

Reference Functions

- OFFSET
- INDIRECT
- INDEX
- MATCH
- We will have detailed discussion on these functions as we go along.

Errors

Errors

- An error refers to a situation where a formula, a function or an operation cannot produce valid result
- This is due to various reasons such as
  - Incorrect data
  - Incorrect syntax
  - Unexpected condition
- Excel displays the error message in the cell where the formula or function is located.
- Errors are represented by special error codes, such as "#DIV/0!", "#VALUE!", "#NAME?"
- These error codes provide information about the nature of the error, making it easier for us to identify and troubleshoot the problem.

Errors

- **Pound # error class**
    - **#VALUE!**
    - **#REF!**
    - **#NUM**
    - **#DIV/0!**

#VALUE! Error Example

#REF! Error Example

#REF! Error

- In cell A6 and cell A7, record the values 10, 20
- In cell B6, perform the addition operation A6 + A7
- Delete row 7
- We get #REF! Error. It states that cell A7 is incorrect reference

Thank You!